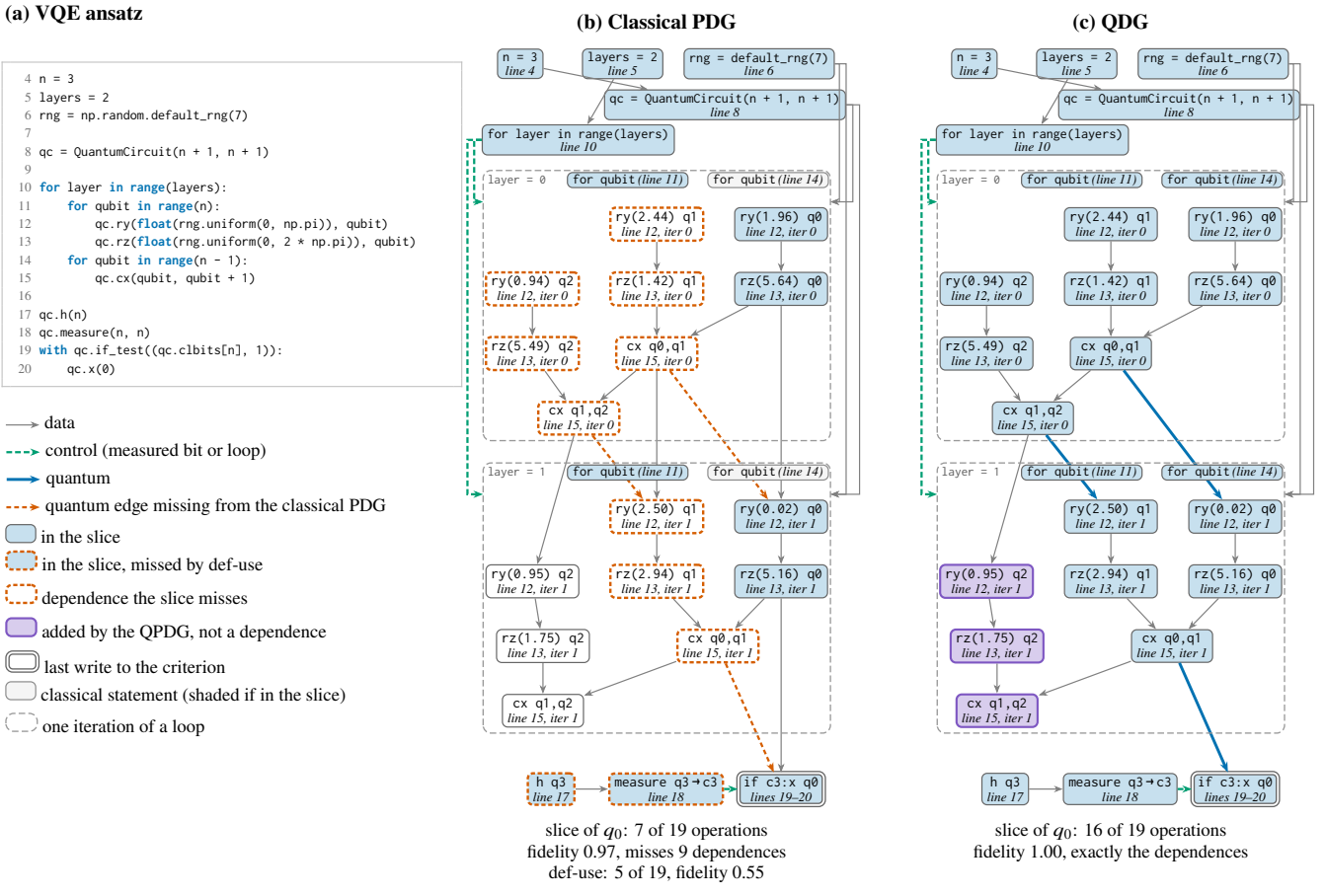

(a) VQE ansatz
4 n = 3
5 layers = 2
6 rng = np.random.default_rng(7)
7
8 qc = QuantumCircuit(n + 1, n + 1)
9
10 for layer in range(layers):
11 for qubit in range(n):
12 qc.ry(float(rng.uniform(0, np.pi)), qubit)
13 qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14 for qubit in range(n - 1):
15 qc.cx(qubit, qubit + 1)
16
17 qc.h(n)
18 qc.measure(n, n)
19 with qc.if_test((qc.clbits[n], 1)):
20 qc.x(0)
data
control (measured bit or loop)
quantum
quantum edge missing from the classical PDG
in the slice
in the slice, missed by def-use
dependence the slice misses
added by the QPDG, not a dependence
last write to the criterion
classical statement (shaded if in the slice)
one iteration of a loop
(b) Classical PDG
n = 3 line 4
layers = 2 line 5
rng = default_rng(7) line 6
qc = QuantumCircuit(n + 1, n + 1) line 8
for layer in range(layers) line 10
layer = 0
for qubit (line 11)
for qubit (line 14)
ry(2.44) q1 line 12, iter 0
ry(1.96) q0 line 12, iter 0
ry(0.94) q2 line 12, iter 0
rz(1.42) q1 line 13, iter 0
rz(5.64) q0 line 13, iter 0
rz(5.49) q2 line 13, iter 0
cx q0,q1 line 15, iter 0
cx q1,q2 line 15, iter 0
layer = 1
for qubit (line 11)
for qubit (line 14)
ry(2.50) q1 line 12, iter 1
ry(0.02) q0 line 12, iter 1
ry(0.95) q2 line 12, iter 1
rz(2.94) q1 line 13, iter 1
rz(5.16) q0 line 13, iter 1
rz(1.75) q2 line 13, iter 1
cx q0,q1 line 15, iter 1
cx q1,q2 line 15, iter 1
h q3 line 17
measure q3→c3 line 18
if c3:x q0 lines 19–20
slice of q_0: 7 of 19 operations
fidelity 0.97, misses 9 dependences
def-use: 5 of 19, fidelity 0.55
(c) QDG
n = 3 line 4
layers = 2 line 5
rng = default_rng(7) line 6
qc = QuantumCircuit(n + 1, n + 1) line 8
for layer in range(layers) line 10
layer = 0
for qubit (line 11)
for qubit (line 14)
ry(2.44) q1 line 12, iter 0
ry(1.96) q0 line 12, iter 0
ry(0.94) q2 line 12, iter 0
rz(1.42) q1 line 13, iter 0
rz(5.64) q0 line 13, iter 0
rz(5.49) q2 line 13, iter 0
cx q0,q1 line 15, iter 0
cx q1,q2 line 15, iter 0
layer = 1
for qubit (line 11)
for qubit (line 14)
ry(2.50) q1 line 12, iter 1
ry(0.02) q0 line 12, iter 1
ry(0.95) q2 line 12, iter 1
rz(2.94) q1 line 13, iter 1
rz(5.16) q0 line 13, iter 1
rz(1.75) q2 line 13, iter 1
cx q0,q1 line 15, iter 1
cx q1,q2 line 15, iter 1
h q3 line 17
measure q3→c3 line 18
if c3:x q0 lines 19–20
slice of q_0: 16 of 19 operations
fidelity 1.00, exactly the dependences